

Prime Factorisation, Divisors & Perfect Powers

Integrated Assimilation Book

Prime factorisation is a coordinate system for integers. Translate the condition into exponent conditions before you count or search.

1. Reconnect: factors are easier in prime coordinates

You already know that $72=8*9=2^3*3^2$. The important upgrade is not the factorisation itself; it is what the exponents remember.

For a positive integer
 $n=p_1^{a_1} p_2^{a_2} \dots p_r^{a_r}$,
 every positive divisor has the form
 $d=p_1^{b_1} \dots p_r^{b_r}$
 with $0 \leq b_i \leq a_i$.

This one statement turns a divisor problem into a finite choice problem.

Fundamental Theorem of Arithmetic

Every positive integer greater than 1 has a prime factorisation, unique apart from the order of the primes. That uniqueness is why exponent arguments are legitimate: two positive integers are equal exactly when the exponent of every prime agrees.

2. Divisibility becomes componentwise comparison

If
 $A=2^5*3^2*7$ and $B=2^8*3^2*5*7^4$,
 then $A|B$: each exponent required by A fits inside the corresponding exponent of B.
 The extra factor 5 in B is harmless.

General rule:

$A|B$ iff $v_p(A) \leq v_p(B)$ for every prime p .

The symbol $v_p(N)$ means the exponent of p in the prime factorisation of N .

3. Count divisors without listing them

For $n=p_1^{a_1} \dots p_r^{a_r}$, each divisor independently chooses exponent $0, 1, \dots, a_i$.
 Therefore

$$\tau(n) = (a_1+1)(a_2+1) \dots (a_r+1).$$

Example: $360=2^3*3^2*5$, so $\tau(360)=4*3*2=24$.

Square divisors

A square divisor must use even exponents. The number of square divisors is
 $(\text{floor}(a_1/2)+1) \dots (\text{floor}(a_r/2)+1)$.

Why divisor parity detects squares

Divisors usually pair as d and n/d . Exactly one divisor can be unpaired: $\sqrt{n}$.

Thus $\tau(n)$ is odd exactly when n is a perfect square.

This is a structural test; enumeration is the wrong route.

4. Perfect powers are exponent divisibility

n is a square iff every prime exponent is even.

n is a cube iff every prime exponent is divisible by 3.

More generally, n is a perfect k -th power iff every prime exponent is divisible by k .

Hence, if $n=2^{18}3^{12}5^6$, the largest $k>1$ for which n is a perfect k -th power is

$\gcd(18,12,6)=6$.

Minimal multiplier or divisor

To make $756=2^2 \cdot 3^3 \cdot 7$ a square, fix only the odd exponents:
multiply by $3 \cdot 7=21$.

To turn a number into a cube, raise each exponent to the next multiple of 3.

5. Squarefree structure

A positive integer is squarefree when no prime square divides it. In prime coordinates, every exponent is 0 or 1.

This is nearly the opposite of a square:

- square: every exponent is even;
- squarefree: every exponent is at most 1.

If a squarefree number has r distinct prime factors, it has exactly 2^r divisors, because each prime is either included or not included.

6. Valuations: focus on one prime at a time

For products, valuations add:

$$v_p(AB)=v_p(A)+v_p(B).$$

For factorials, Legendre's formula gives

$$v_p(n!)=\text{floor}(n/p)+\text{floor}(n/p^2)+\text{floor}(n/p^3)+\dots$$

Example:

$$v_2(25!)=12+6+3+1=22.$$

This is much cheaper than calculating $25!$.

A key decision boundary

If the question asks whether a candidate divides $n!$, compare each required prime exponent with the corresponding valuation of $n!$.

If it asks only for a gcd or lcm of two ordinary integers, the earlier gcd/lcm machinery may be cheaper than full prime factorisation.

7. Factor pairs and coprimality

When $xy=N$, the exponent vector of N is split between x and y .

If no gcd condition is present, each prime exponent may be divided between the two factors in many ways.

If $\gcd(x,y)=1$, the rule becomes much sharper: for every prime power $p^a \parallel N$, the whole block p^a must go to exactly one of x,y . Splitting the exponent would put p in both factors.

So if N has r distinct prime factors, the number of ordered coprime factor pairs (x,y) with $xy=N$ is 2^r .

8. Extremal reconstruction from a divisor count

Suppose $\tau(n)=12$. Factor 12 as exponent-choice products:

- 12 $\rightarrow$ exponent pattern 11;
- $6 \cdot 2 \rightarrow$ pattern 5,1;
- $4 \cdot 3 \rightarrow$ pattern 3,2;
- $3 \cdot 2 \cdot 2 \rightarrow$ pattern 2,1,1.

To make n as small as possible, put larger exponents on smaller primes. Candidates are 2^{11} , $2^5 \cdot 3$, $2^3 \cdot 3^2$, $2^2 \cdot 3^5$.

The minimum is 60.

This is the governing extremal principle:

For a fixed multiset of positive exponents, assign them in non-increasing order to primes in increasing order.

9. Historical decision models

Two perfect squares separated by 13

A fixed difference between squares suggests difference-of-squares factorisation before any age enumeration. The factor pair of 13 fixes the ages immediately.

Smallest integer not dividing a factorial

The surface says "smallest integer", but the hidden test is prime-exponent capacity in the factorial. This is valuation structure, not a long divisibility table.

Distinct-square average

Averages first determine how many squares and their total. Then the extremal question becomes "how much total must be reserved for the smaller distinct squares?"

Divisors of n^2 below n

Divisor pairing around $\sqrt{n^2}=n$ halves the divisor set before any detailed classification.

Cumulative divisor parity

Replace every $d(i)$ by one bit: odd for squares, even otherwise. The enormous-looking sum becomes a count of square indices.

10. Diagnose the common failures

- Factor-list reflex: listing divisors when exponent choices count them immediately.
- Prime-set confusion: forgetting multiplicity; 2^5 and 2 have the same prime set but very different divisor structure.
- Square/squarefree reversal: "even exponents" and "exponents at most one" are not interchangeable.
- Perfect-power guesswork: checking numerical roots instead of taking gcd of exponents.
- Factorial expansion: calculating a factorial instead of a valuation.
- Coprime-pair splitting error: splitting a prime power between both factors despite $\gcd(x,y)=1$.
- Extremal exponent order error: assigning the largest exponent to a larger prime.

11. Fade toward independence

Full support

Find the number of positive divisors of $2^4 \cdot 3^2 \cdot 5$.

Write one factor for each independent exponent choice.

Reduced support

What is the least multiplier that makes $2^5 \cdot 3^2 \cdot 7^3$ a square?

Light support

A number has exactly 18 positive divisors. What exponent patterns are possible?

Independent

Find the smallest positive integer with exactly 45 positive divisors.

Do not search integer-by-integer. Decide what the divisor-count factorisations force.

12. Adopt the router

When an integer-structure problem appears, ask:

1. Can I write or infer its prime exponent vector?
2. Is the target about divisibility, a divisor count, a perfect power, squarefreeness, a valuation, or a factor pair?
3. What condition does that target impose on each exponent?
4. Are exponent choices independent, complementary, or forced into whole blocks?
5. If minimizing, have I assigned the largest exponents to the smallest primes?
6. Have I checked all prime exponents and all boundary cases?

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First-Step Reference

One-line router

FACTOR / INFER EXPONENTS -> TRANSLATE TARGET -> COUNT OR RECONSTRUCT -> CHECK

Visible clue	First mathematical line
"number of divisors"	write $n = \prod p_i^{a_i}$ and $\tau(n) = \prod (a_i + 1)$
"perfect square"	require every a_i even
"perfect cube"	require every a_i divisible by 3
"greatest perfect-power exponent"	compute $\gcd(a_1, \dots, a_r)$
"squarefree"	require every exponent 0 or 1
"divides $n!$ "	compare $v_p(\text{candidate})$ with $v_p(n!)$
"factor pairs"	pair complementary exponent choices
" $xy=N$, $\gcd(x,y)=1$ "	assign each complete prime-power block to one side
" $\tau(n)$ odd"	test whether n is a square
"smallest n with $\tau(n)=K$ "	factor K into (a_i+1) patterns
"divisors of n^2 below n "	pair divisors around n first

Close contrasts

- gcd/lcm target -> retrieve gcd/lcm structure; multiplicity target -> exponent vector.
- square -> all exponents even; squarefree -> all exponents at most 1.
- count divisors -> independent exponent choices; count factor pairs -> complementary choices.
- candidate divides factorial -> valuation; last digits of factorial -> modular machinery elsewhere.
- minimize number with fixed τ -> exponent pattern + ordered primes; merely compute τ -> no minimization.

Checks

1. Did every prime appearing in the number appear in the vector?
2. Did you accidentally use a square criterion for a cube or vice versa?
3. For $\gcd(x,y)=1$, did any prime enter both factors?
4. For an extremal construction, did a larger exponent land on a larger prime?
5. For historical source work, did you use the validated statement and key status?

Recognition and First-Line Lab

Write only the first useful mathematical object or line unless the item asks for a value.

1. $N=2^7 \cdot 3^4 \cdot 5^3$. First line for the number of square divisors.
2. 756 must be multiplied by the least positive integer to become a square. First line.
3. $M=2^{18} \cdot 3^{12} \cdot 5^6$. First line for the greatest k such that M is a perfect k -th power.
4. Decide whether 1764 has an odd or even number of divisors without listing them.
5. First line for $v_3(50!)$.
6. If $xy=2^6 \cdot 3^4$ and $\gcd(x,y)=1$, state the structural restriction on the factors.
7. A positive integer has exactly 12 divisors. List the possible exponent patterns.
8. Explain in one sentence why divisors of n^2 pair around n .
9. Which route is cheaper for $\gcd(987654, 123456)$: Euclid or full prime factorisation?
10. Which route is correct for "largest k such that $10^{24} \cdot 21^{12}$ is a perfect k -th power"?

Practice and Transfer Bank

Practice

- Factor 7560 into primes and write its exponent vector.
- Find the number of positive divisors of 3600.
- Find the number of square divisors of $2^7 \cdot 3^4 \cdot 5^3$.
- Find the least positive integer by which 756 must be multiplied to make a perfect square.
- Find the least positive integer by which 5400 must be divided to leave a perfect square.
- Find the greatest $k > 1$ for which $2^6 \cdot 3^3 \cdot 5^9$ is a perfect k -th power.
- Find the squarefree kernel of 7560, meaning the product of primes whose exponents in 7560 are odd.
- Without listing divisors, decide whether 1764 has an odd number of positive divisors.
- Find the number of unordered positive factor pairs of 360.
- A squarefree integer has five distinct prime factors. How many positive divisors does it have?
- Compute $v_2(25!)$.
- Compute $v_3(50!)$.
- Find the greatest k for which $2^{18} \cdot 3^{12} \cdot 5^6$ is a perfect k -th power.
- How many square divisors does $2^8 \cdot 3^5 \cdot 7^2$ have?
- Find the smallest positive integer with exactly 12 positive divisors.
- Find the smallest positive integer with exactly 15 positive divisors.
- Find the smallest positive integer with exactly 45 positive divisors.
- How many cube divisors does $2^{10} \cdot 3^6$ have?
- How many ordered pairs (x, y) of positive coprime integers satisfy $xy = 2^6 \cdot 3^4$?
- How many squarefree positive integers divide $2^4 \cdot 3^2 \cdot 5^7 \cdot 7$?
- How many integers $n \leq 500$ have an odd number of positive divisors?
- Let $N = 2^4 \cdot 3^2$. How many positive divisors of N^2 are less than N but do not divide N ?
- Find the least multiplier that makes $2^5 \cdot 3^8 \cdot 5$ a perfect sixth power.
- A divisor of $2^5 \cdot 3^4 \cdot 5^3 \cdot 7$ is squarefree and has exactly three prime factors. How many such divisors are there?

Transfer

- A rectangular box has integer side lengths with product $2^6 \cdot 3^3$. If the three side lengths are pairwise coprime, how many ordered triples of side lengths are possible?
- A locker code is the exponent vector (a, b, c) of a divisor of $2^8 \cdot 3^5 \cdot 5^4$. How many codes correspond to square divisors?
- The integer m is known to divide $30!$ and has prime factorisation $2^a \cdot 3^b \cdot 5^c$. State the sharp independent bounds on a, b, c .
- Explain why "the number of divisors is odd" is a better recognition clue than "list the divisors" for detecting a square.
- Two positive squares differ by 45. Give all possible ordered pairs (larger square root, smaller square root).
- A positive integer has exponent vector $(8, 4, 2)$. Which of square, cube, fourth power, sixth power must it be?

Independent Mastery Check

No method is named. Choose your representation and justify each answer.

1. Find the smallest positive integer with exactly 24 positive divisors.
2. How many positive divisors of $2^9 \cdot 3^6 \cdot 5^2$ are perfect squares?
3. Find the least positive multiplier that turns $2^7 \cdot 3^2 \cdot 5^5$ into a perfect cube.
4. Find $v_5(100!)$.
5. How many positive integers at most 1000 have an odd number of positive divisors?
6. Find the greatest $k > 1$ such that $2^{30} \cdot 3^{18} \cdot 7^{12}$ is a perfect k -th power.
7. Let $N = 2^6 \cdot 3^4 \cdot 5^2$. How many ordered coprime factor pairs (x, y) satisfy $xy = N$?
8. Find the smallest squarefree positive integer with exactly 32 positive divisors.
9. A divisor of $2^6 \cdot 3^5 \cdot 5^4 \cdot 7^3 \cdot 11^2$ is squarefree and uses exactly three primes. How many such divisors are possible?
10. Let $N = 2^5 \cdot 3^3$. How many divisors of N^2 are less than N but do not divide N ?
11. Prove or refute: a positive integer has an even number of positive divisors exactly when it is not a square.
12. Positive integer squares $A^2 > B^2$ differ by 21. Determine all possible pairs (A, B) .